

Agent P&L and Portfolio Diversity

Outcome Analysis by Slider Settings

Data: January 8-11, 2026 | Agents with >10 ETH Start

Generated: 2026-01-10 20:00

*Note: P&L based on ETH balance change only.
Does not account for token holdings value.*

EXECUTIVE SUMMARY: P&L AND DIVERSITY ANALYSIS

SAMPLE: 27 agents with >10 ETH starting balance (Jan 8+ data)

KEY FINDINGS:

1. RISK PREFERENCE IS THE STRONGEST P&L PREDICTOR

Correlation with ETH Change %: $r = -0.514$ ($p = 0.006$) ***

- Higher risk preference = WORSE returns
- Risk level 5: -28.2% avg P&L
- Risk level 1: 0.0% avg P&L (no trading, no loss)

2. TRADING ACTIVITY VALIDATES AS EXPECTED

Correlation with Total Trades: $r = +0.491$ ($p = 0.009$) ***

- Higher activity setting = More trades executed
- Setting works as intended

3. DIVERSIFICATION SETTING VALIDATES

Correlation with Unique Tokens: $r = +0.472$ ($p = 0.013$) **

- Higher diversification = More unique tokens traded
- But also correlates with worse P&L ($r = -0.37$, $p = 0.06$)

4. NO AGENTS HAD POSITIVE P&L

Distribution:

- 17 agents (63%): Flat (0% change)
- 4 agents (15%): Small Loss (-1% to -10%)
- 6 agents (22%): Big Loss (>-10%)
- 0 agents: Any gains

5. LOSERS PROFILE

Big Loss agents had higher:

- Risk preference (4.0 vs 2.2 avg)
- More trades (2.8 vs 0.3 avg)
- More diversification (3.3 vs 2.1 avg)

INTERPRETATION:

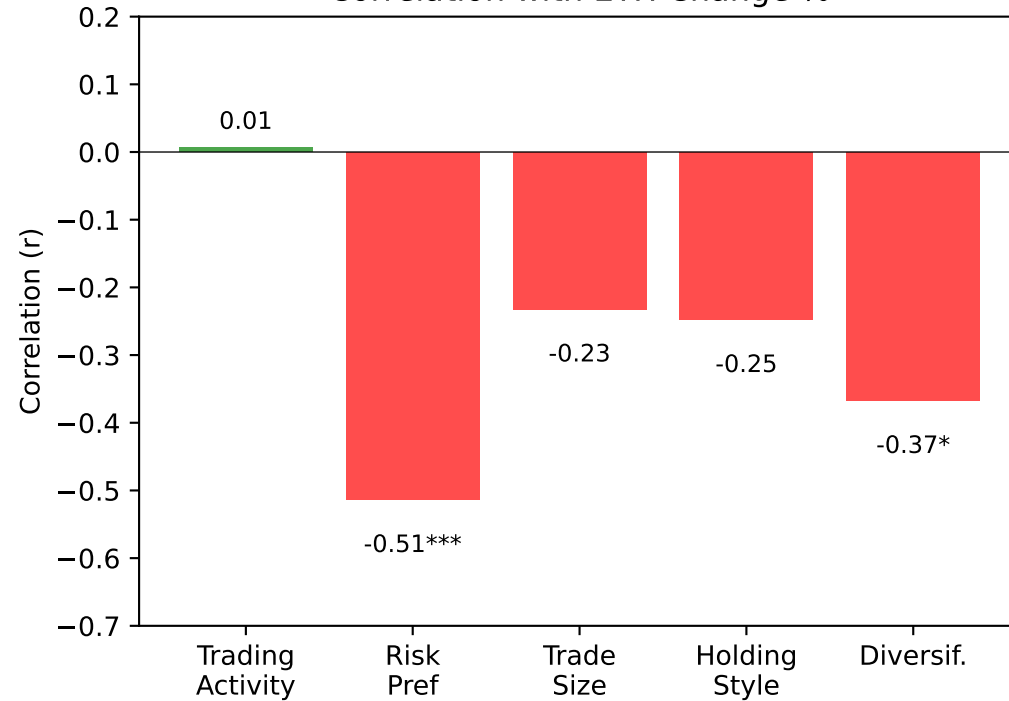
In this market period, the winning strategy was:

- LOW risk preference (don't trade)
- LOW trading activity
- LOW diversification

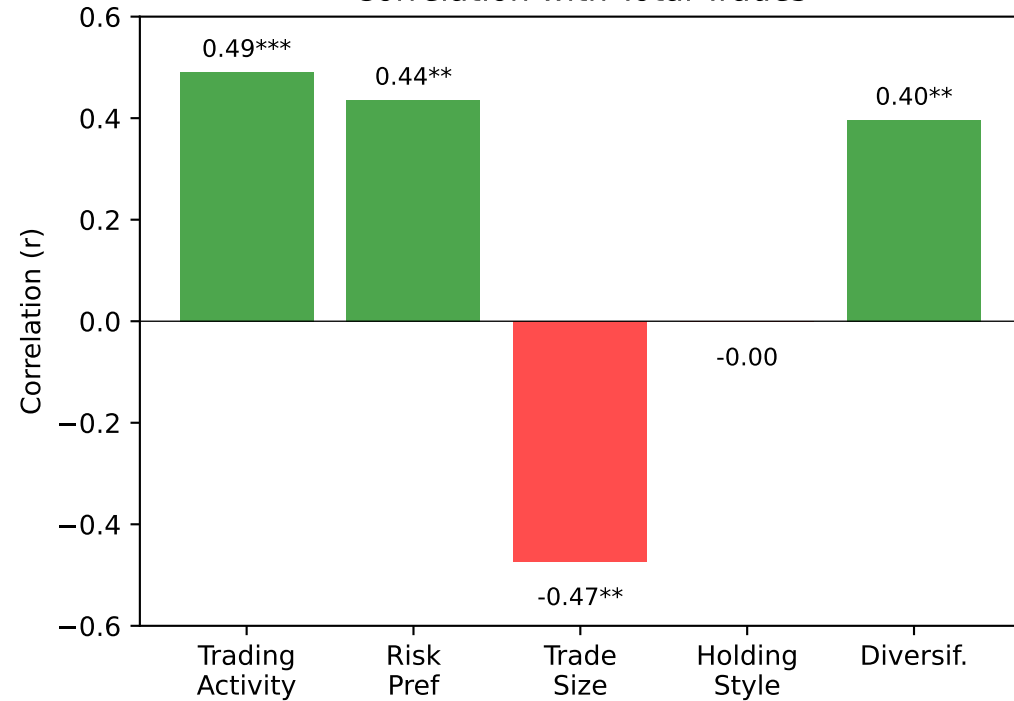
Trading activity led to losses, regardless of configuration.

Correlation Analysis: Settings vs Outcomes

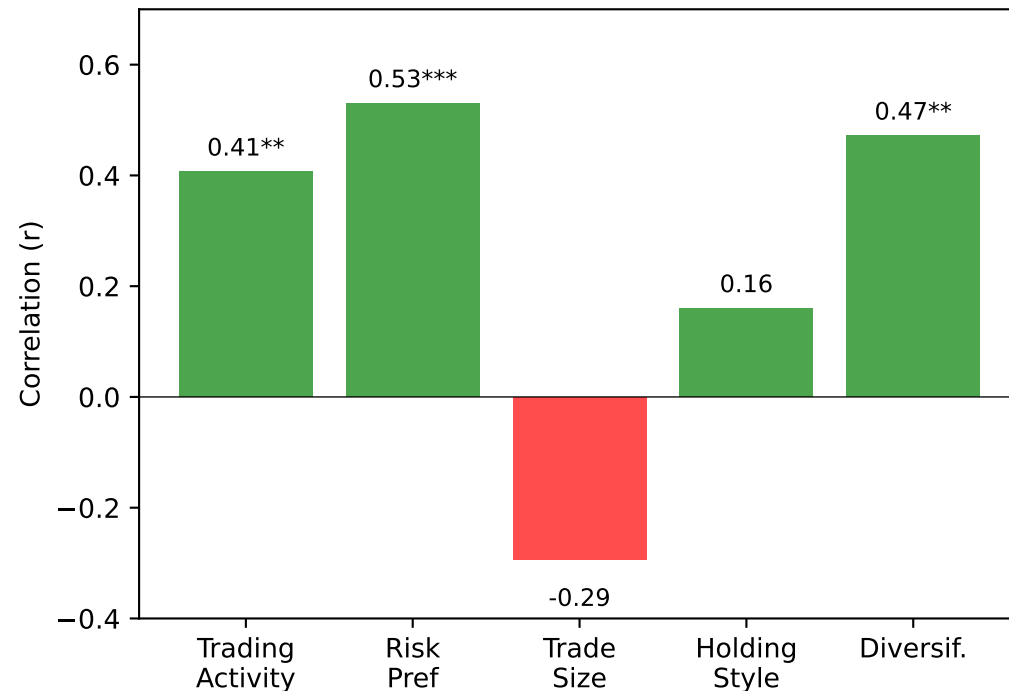
Correlation with ETH Change %



Correlation with Total Trades



Correlation with Unique Tokens Traded



STATISTICAL SIGNIFICANCE:

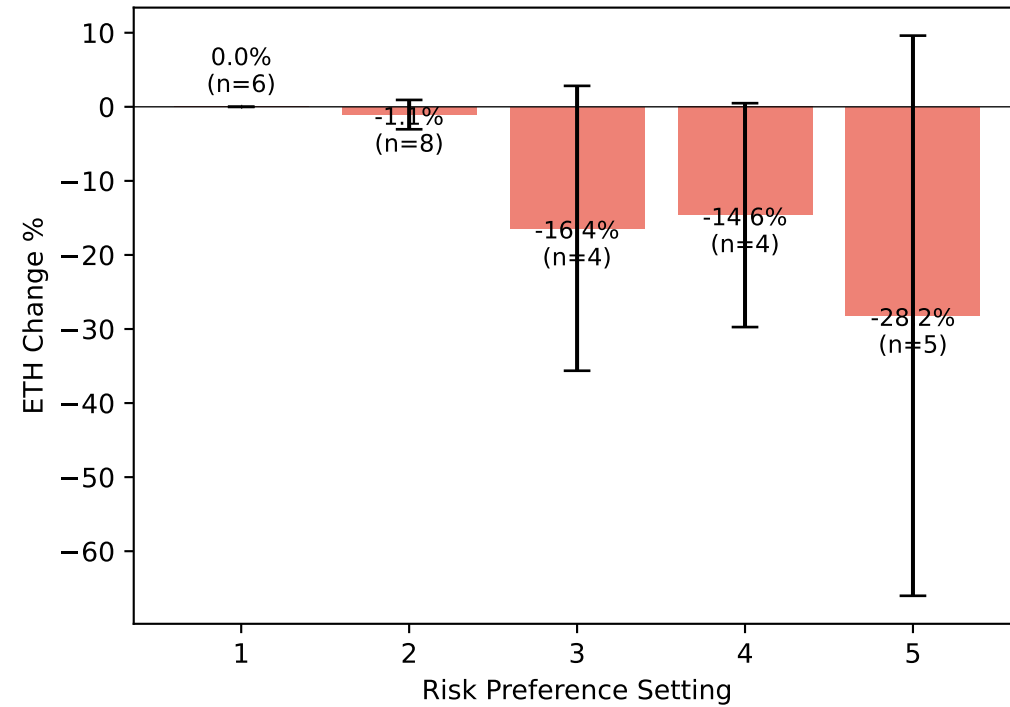
- *** $p < 0.01$ (highly significant)
- ** $p < 0.05$ (significant)
- * $p < 0.10$ (marginally significant)

INTERPRETATION:

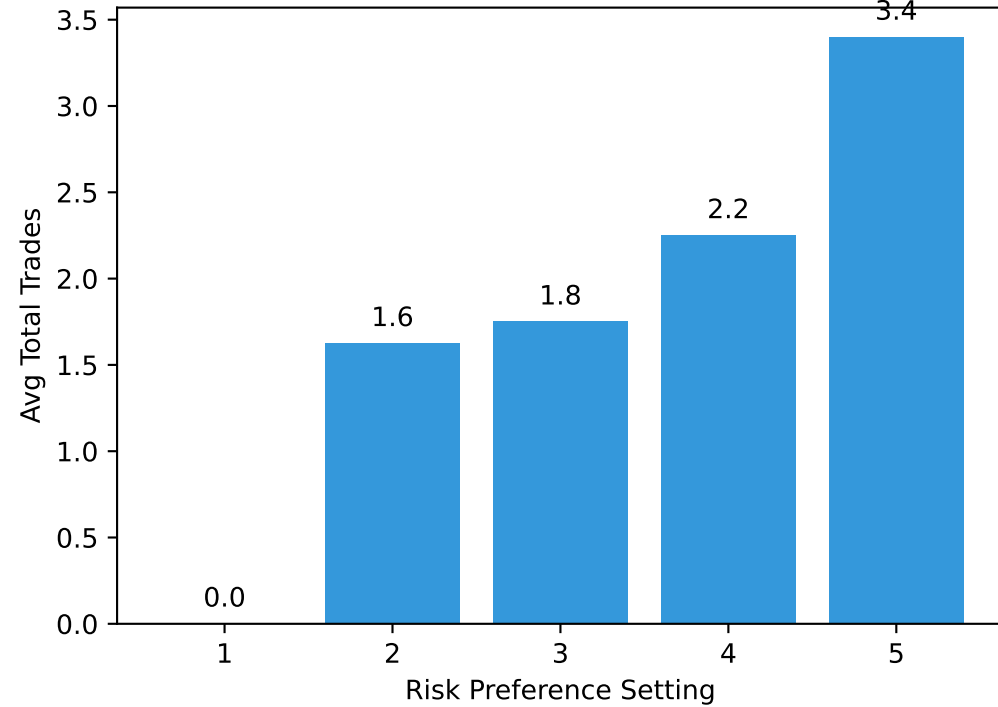
- Risk Preference is the only slider significantly correlated with P&L ($r = -0.51$, $p = 0.006$)
- Trading Activity, Risk Preference, and Diversification all significantly correlate with trading behavior
- Trade Size negatively correlates with trades (counterintuitive)

Risk Preference: The Key P&L Driver

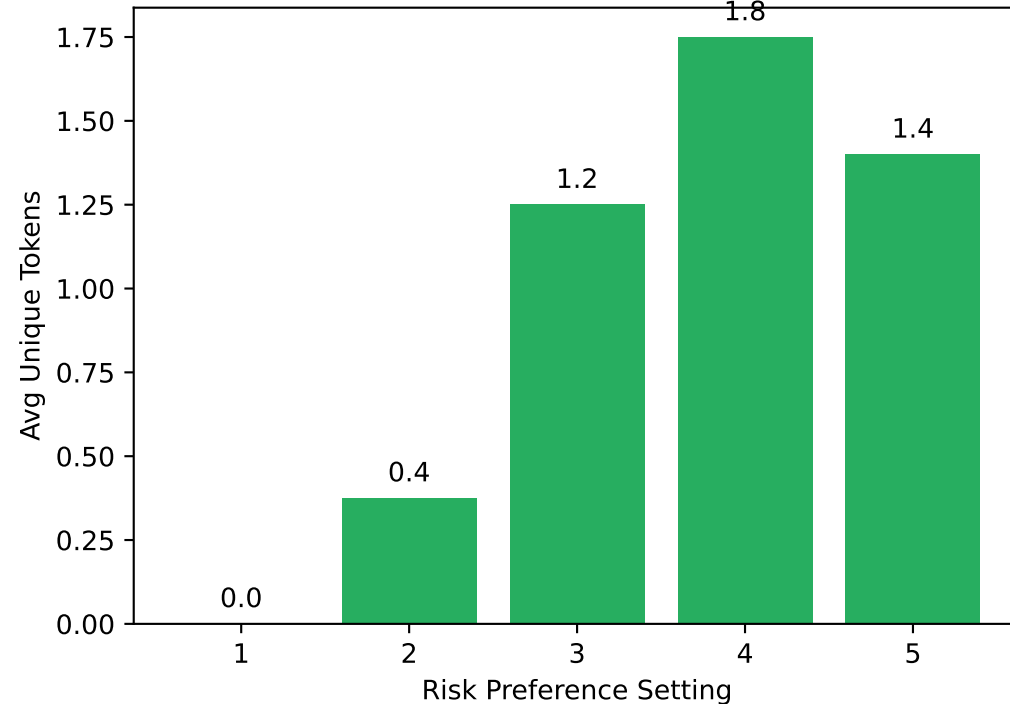
Average P&L by Risk Preference



Trading Activity by Risk Preference



Token Diversity by Risk Preference



RISK PREFERENCE IMPACT:

Setting 1 (Risk Averse):

- 0.0% avg P&L
- 0.0 avg trades
- Don't trade = Don't lose

Setting 5 (Risk Seeking):

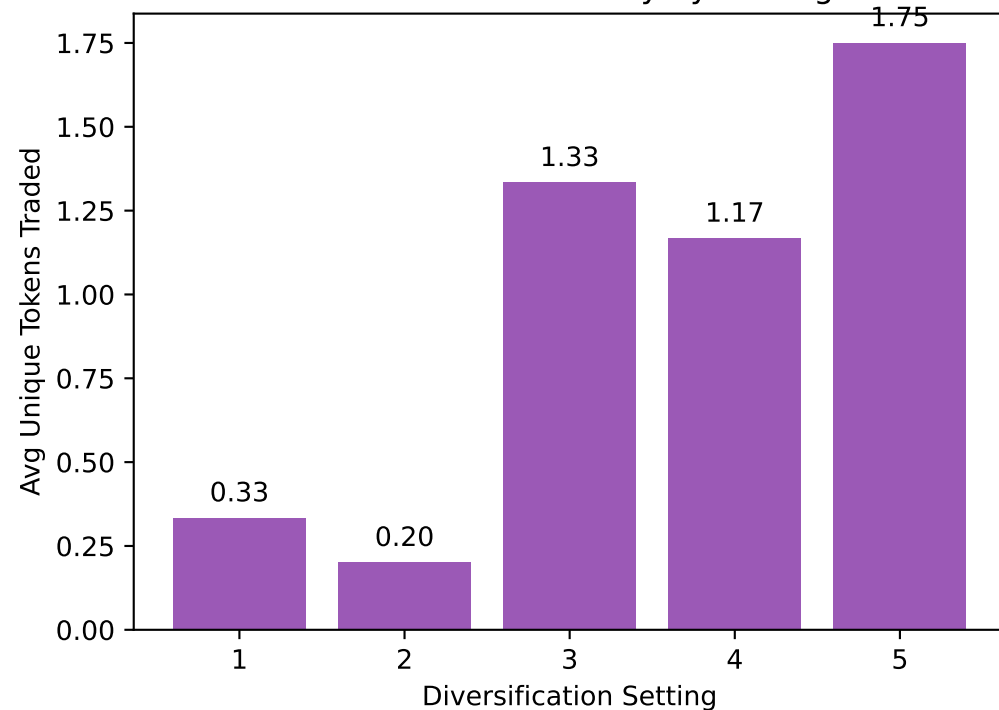
- -28.2% avg P&L
- 3.4 avg trades
- High activity = High losses

TAKEAWAY:

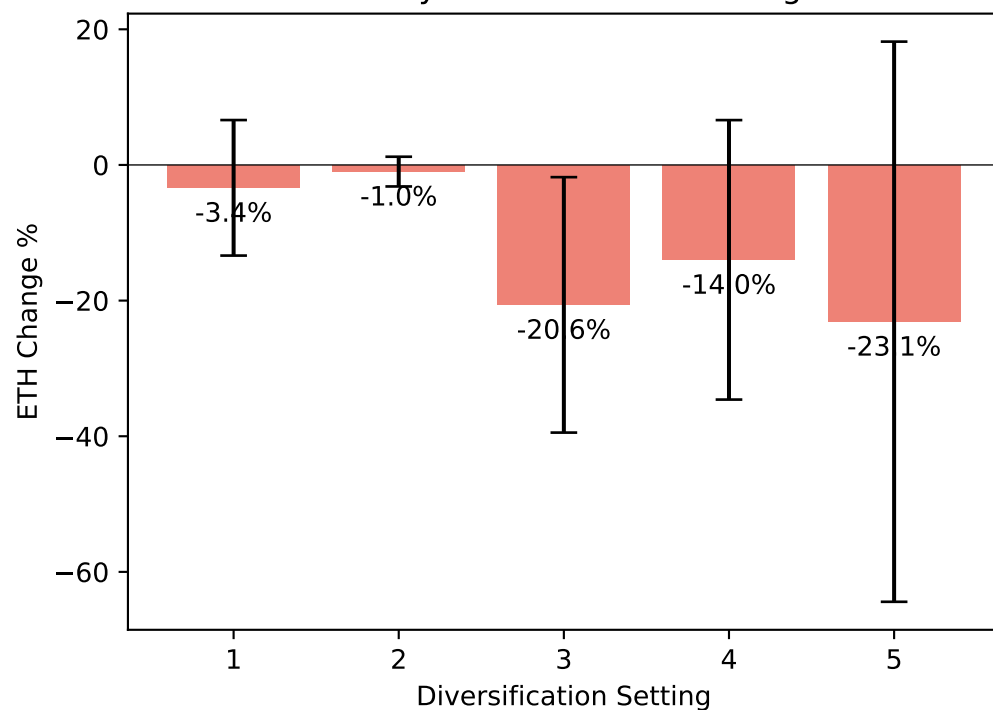
In this market period, risk-seeking behavior was punished. Conservative agents preserved capital by not trading.

Diversification Setting: Behavior Validation

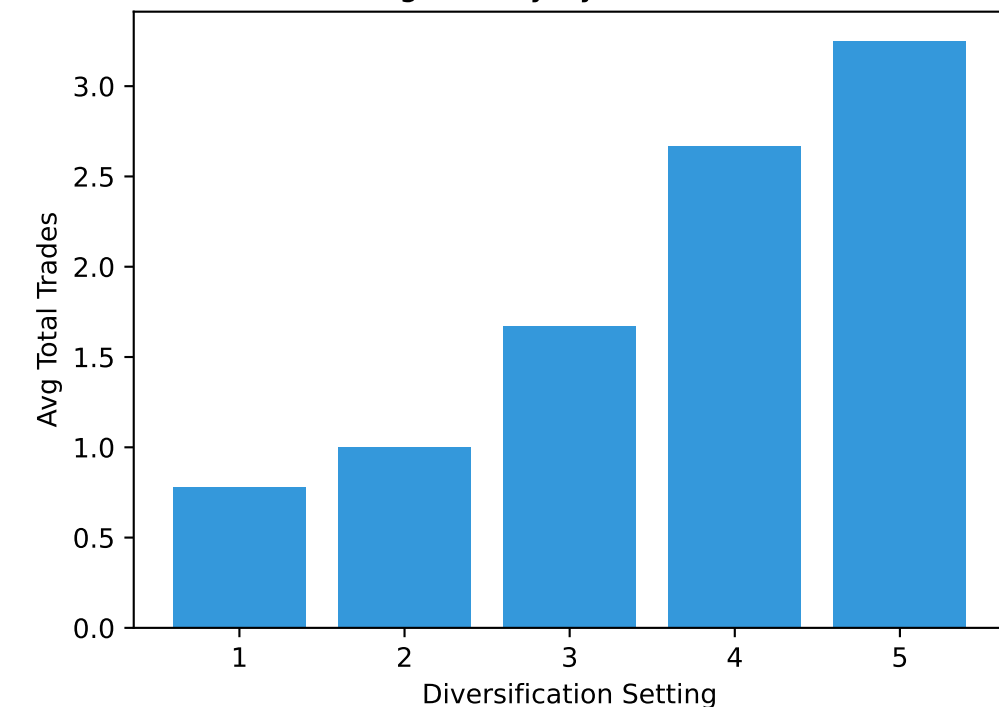
Actual Token Diversity by Setting



P&L by Diversification Setting



Trading Activity by Diversification



DIVERSIFICATION VALIDATION:

Setting validates (higher = more tokens):

- Setting 1: 0.33 unique tokens
- Setting 5: 1.75 unique tokens

BUT higher diversification also led to worse P&L in this period:

- Setting 1: -3.4% avg
- Setting 5: -23.1% avg

CORRELATION:

- With Unique Tokens: $r=+0.47$ ($p=0.013$)**
- With P&L: $r=-0.37$ ($p=0.059$)*

More diverse = More trading = More losses
(in this down market)

P&L Category Profile Analysis

P&L CATEGORY DISTRIBUTION

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Category	Count	% of Total
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Big Loss (>-10%)	6	22%
Small Loss (-1 to -10%)	4	15%
Flat (no change)	17	63%
Small Gain	0	0%
Big Gain	0	0%

AVERAGE SETTINGS BY CATEGORY

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	Big Loss	Small Loss	Flat
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Trading Activity	3.17	4.50	2.59
Risk Preference	4.00	3.25	2.24
Trade Size	3.17	1.00	2.71
Holding Style	3.17	2.75	2.59
Diversification	3.33	4.00	2.12
Total Trades	2.83	6.00	0.29
Unique Tokens	1.83	2.25	0.12

KEY DIFFERENCES (Losers vs Flat):

- Higher Risk Preference: 4.0 vs 2.2
- More Trades: 2.8-6.0 vs 0.3
- More Token Diversity: 1.8-2.3 vs 0.1

CONCLUSION:

In this market period, agents who traded actively lost money.
Agents who stayed flat (minimal trading) preserved capital.
The "best" strategy was: don't trade.

Data Limitations & Methodology Notes

DATA LIMITATIONS

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1. SAMPLE SIZE
 - Only 27 agents with valid >10 ETH starting balance
 - Many agents (418/472) have NULL start_eth values
 - Small sample limits statistical power
2. P&L MEASUREMENT
 - Based ONLY on ETH balance change
 - Does NOT account for token holdings value
 - True P&L would require token price data at start/end
 - Current metric is a PROXY, not actual profit/loss
3. TIME PERIOD
 - Only 4 days of data (Jan 8-11, 2026)
 - Market conditions during this period may not be representative
 - Longer time series needed for robust conclusions
4. CONFOUNDING FACTORS
 - Strategy presence not fully captured in cross-analysis
 - Market conditions varied during the period
 - Agent entry times vary

METHODOLOGY

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1. Starting Balance: First snapshot ETH balance per agent
2. Ending Balance: Last snapshot ETH balance per agent
3. P&L %: $(\text{End} - \text{Start}) / \text{Start} * 100$
4. Correlations: Pearson correlation with p-values
5. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$

RECOMMENDATIONS FOR FUTURE ANALYSIS

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1. Extend time period for more observations
2. Incorporate token holdings value in P&L calculation
3. Control for market conditions (up vs down markets)
4. Larger sample with valid starting balances